

Tools and the night sky

READINGS AND SOURCES

Name: _____

Date: _____

READING A / How telescopes help us

A telescope gathers light from distant objects. Its lenses or mirrors focus that light to make an image. Larger lenses or mirrors can collect more light, helping astronomers observe faint objects.

People can use a telescope to study details that their eyes alone cannot easily see. Choosing a viewing tool depends on what you want to observe.

Source card: Publisher: NASA. Website: NASA Space Place. Format: article.

Original source: How Do Telescopes Work?

spaceplace.nasa.gov/telescopes/en/

READING B / Finding a place to skywatch

You can begin skywatching with your eyes. Binoculars and telescopes reveal finer details.

Light from streets and buildings can scatter into the night sky. This glow makes faint stars harder to see. A darker, safe viewing place can help.

Source card: Publisher: NASA. Website: NASA Science. Format: questions and answers.

Original source: Skywatching FAQ

science.nasa.gov/skywatching/faq/

SKYWATCHING REMINDER

Observe with an adult and choose a safe location. Never look directly at the Sun or point binoculars or a telescope at it.

Know, wonder, learn

BEFORE AND AFTER READING

Name: _____

Date: _____

How do objects in space impact daily life? Add ideas now, then return after reading.

KNOW
What do I already know?

I think...

WONDER
What am I curious about?

I wonder...

LEARN
What did I learn?

I learned... My evidence is...

Revisit an idea
What did the evidence confirm or change?

At first I thought... Now I think...

Why can we see more stars?

MAKING AN INFERENCE

Name: _____ Date: _____

Two groups observe on the same clear night. The group at a dark park records more stars.

The text says
Write a useful detail from Reading B on page 1.

The text says...

I already know
What helpful idea do you bring to this observation?

I know...

I can infer
Combine the reading detail and your knowledge.

I infer... because...

Explain why more stars were visible. Support your idea with evidence.

A place for skywatching

PROS AND CONS

Name: _____

Date: _____

A class can use its lit schoolyard or travel 30 minutes to a darker park. Both are safe. The trip needs transport.

Advantages

How could the darker park help?

The park could help us... Use a detail from Reading B.

Challenges

What could make the trip difficult?

The trip could make it harder to... Use the scenario.

My decision

Which place would you choose, and why?

I would choose... because...

Consider both the viewing conditions and the practical needs of the class.

A brighter sky, fewer stars

PROBLEM AND SOLUTION

Name: _____ Date: _____

Nearby floodlights point upward. A class wants to see faint stars while keeping walkways safely lit.

The problem
What is getting in the way? Use Reading B as evidence.

Our problem is...

A possible solution
What could the class propose? Include safe lighting for people.

Our plan is... It could help because...

Test the plan
What would you observe before and after the change?

I would compare... I would keep... the same.

Try to compare observations under similar sky conditions.

Who made this source?

ACAPS / SOURCE ANALYSIS

Name: _____

Date: _____

Use Reading A and its source card on page 1. Support each response with a detail. You can also explore the linked original article.

A / Author Who created the original source?	My response and evidence: _____ _____
C / Context On which website does it appear? What kind of text is it?	My response and evidence: _____ _____
A / Audience Who might use this information? What is your clue?	My response and evidence: _____ _____
P / Purpose What does it help readers understand?	My response and evidence: _____ _____
S / Significance How does it help our space inquiry?	My response and evidence: _____ _____

For audience and purpose, explain which reading details support your idea.

The voice of a source

SOAPSTONE / SOURCE ANALYSIS

Name: _____

Date: _____

Use Reading B and its source card on page 1. Think about the information, the reader and the way the passage is written.

Speaker Whose original information is this passage based on?	My response and evidence: _____ _____
Occasion What skywatching need or question does it address?	My response and evidence: _____ _____
Audience Who might find this information useful?	My response and evidence: _____ _____
Purpose What can readers learn or do?	My response and evidence: _____ _____
Subject What is the main topic?	My response and evidence: _____ _____
Tone How does the reading sound? Give a text clue.	My response and evidence: _____ _____

Tone can be shown through explanations, suggestions or direct questions.

Space words at work

EIGHT PARTS OF SPEECH

Name: _____ Date: _____

Find one example of each part of speech in the sentence. Write it in the matching box.

Wow! We carefully observe bright stars in the sky, and they shine.

Noun
Names a person, place or thing.

Pronoun
Takes the place of a noun.

Verb
Shows an action or a state of being.

Adjective
Describes a noun.

Adverb
Here, tells how an action happens.

Preposition
Shows a relationship, such as place.

Conjunction
Joins words or parts of a sentence.

Interjection
Expresses a sudden feeling.

Explain a word's job
Choose one word. What does it do in this sentence?
